

Circuit Simulation Lab:10

D Flip-Flop

Introduction

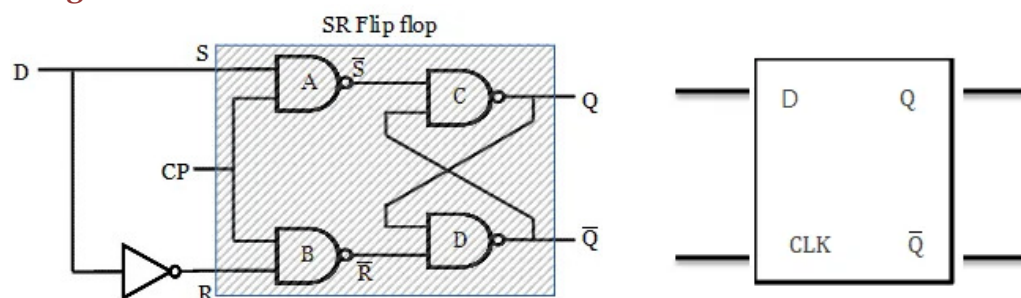
The purpose of this experiment is to introduce the design of simple sequential circuits, in this case D Flip Flop. A D (or Delay) Flip Flop is a digital electronic circuit used to delay the change of state of its output signal (Q) until the next rising edge of a clock timing input signal occurs.

Software tools Requirement

Modelsim (Siemens)

Xilinx Vivado

Logic Diagram



D Flip Flop

D	CLK	$\bar{Q}$
0	Rising Edge	0
1	Rising Edge	1

Truth Table

Describe the above in Verilog HDL and capture the Waveforms

- 1) Dataflow modeling
- 2) Behavior modeling
- 3) Structural modeling (Use the gates for designing the Flop)